

Is a syndromic approach well suited to foodborne disease surveillance? Implication for salmonellosis surveillance and prevention in France from a “One Health” perspective

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ABSTRACT

In response to the increasing emergence of zoonotic pathogens, flexible, multisectoral surveillance systems capable of generating alerts thanks to rapid, nonspecific detection, are crucial before pathogens reach human populations. Syndromic surveillance has proven to be a breakthrough for near real-time disease surveillance in the public health sector. It relies on existing nonspecific data, usually collected for other purposes. *Salmonella* is one of the leading causes of foodborne illness. Data from animal and human sectors can be used to help monitor it across the food chain and thus quickly detect outbreaks. For the first time, this work uses five datasets identified after mapping farm-to-fork surveillance systems in France to apply syndromic surveillance to foodborne *Salmonella* outbreaks. The datasets covered on-farm cattle mortality, laboratory *Salmonella* isolations, and entries to public emergency services from 2011 to 2018. Weekly time series were retrospectively analyzed with five detection algorithms (Holt-Winters, historical limits, exponentially weighted moving average, Shewhart, and cumulative sum) to identify abnormal excess events over the last three years. Our analysis revealed several simultaneous excess events reported across algorithms and datasets from both animal and human health sectors. This shows the potential of syndromic surveillance for monitoring foodborne disease using a One Health approach. Nevertheless, data quality and real-time data collection are key to reliable syndromic surveillance. Temporal and geographic resolutions can also affect anomaly detection. The added value of the different algorithms tested underline the importance of using statistics with complementary characteristics.

1. Introduction

It has been estimated that about one billion cases of illness and millions of deaths occur every year worldwide from zoonoses (Kareesh et al., 2012; Morens and Fauci, 2020). This represents economic losses exceeding hundreds of billions of US dollars in the past 20 years (Smith et al., 2019b). In light of this major health burden, it is critical to implement robust and sensitive surveillance systems able to detect pathogens early enough to serve as an alert.

For many years, disease control in many countries has relied on (i) clinical diagnoses (based on visual observation and lab testing), (ii)

biosecurity measures (e.g., quarantine or culling) and (iii) monitoring trade and livestock movements. However, these traditional approaches are reaching their limits in terms of effectiveness and cost/benefit ratio in the case of infrequent diseases or emerging pathogens. Indeed, passive disease surveillance is widely acknowledged to suffer from under-detection (not all cases are detected), under-diagnosis (not all cases are correctly identified, depending on the quality of lab tests), and under-reporting (not all identified cases are reported). Active surveillance systems are more costly and although they may provide an accurate assessment of the epidemiological situation in the population at a given time, they are not designed for early disease detection.

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Furthermore, these surveillance systems are generally implemented separately in each sector with limited intersectoral collaboration. These limits, associated with the increasing emergence of zoonotic pathogens (Anthony et al., 2013; Murray et al., 2018), underline the need for more flexible surveillance systems capable of rapidly detecting pathogens before they reach human populations. Moreover, the operationalization of a multisectoral, One Health approach, is encouraged by international organizations (World Health Organization (WHO) et al., 2019) to attain global health security.

Syndromic surveillance (SyS) has emerged as a promising tool in animal health (Dorea et al., 2011) and public health (Mandl et al., 2004). It consists in collecting, analyzing, interpreting, and disseminating nonspecific indicators such as clinical signs, which are not necessarily linked to a specific disease. These indicators are typically monitored in real time to detect deviations from normal patterns, thus providing early warning of potential outbreaks or unusual events. The assumption is that the baseline level of a given indicator of population health will change when a hazard-related event occurs, and that such changes can be highlighted in routinely analyzed data (Mandl et al., 2004). SyS is assumed to be more cost-effective than traditional surveillance systems because it uses data collected in another framework. However, because the data sources are unspecific, SyS is typically used to supplement traditional surveillance systems. Despite the promising potential of SyS, in particular related to the increasing digitization of data, there are still gaps in its implementation and evaluation, particularly in the field of animal health. In 2012, a study identified 33 active and pilot SyS systems in public health and 27 in animal health (Dupuy et al., 2013). Gastrointestinal diseases were cited as one of the most frequently monitored syndromes, but very few publications mentioned the existence of One Health systems. These studies focused on the application of SyS in human health or isolated pilot projects in animal health, but few addressed the potential of combining data from SyS and traditional surveillance systems to detect foodborne outbreaks across sectors. The present study seeks to address this gap by evaluating whether it is possible to improve the detection of foodborne outbreaks (particularly those related to *Salmonella*) by combining multiple sources of surveillance data, including nonspecific and disease-related data.

Europe's NOVA project (<https://onehealth.ejp.eu/jrp-nova/>) provided the framework for our study, which focuses on enhancing surveillance systems to better capture outbreaks at the human-animal interface. This One health approach could guide decision making toward more effective management actions (Stark et al., 2015). Despite control measures implemented in food production systems, foodborne infections remain one of the leading causes of illnesses throughout the world (Pires et al., 2015; European Food Safety et al., 2023). The World Health Organization estimated that there were 153 million non-typhoidal *Salmonella* infections worldwide in 2010, of which 56,969 were lethal and nearly half were foodborne (Kirk et al., 2015). In France, *Salmonella* has been identified as the main cause of foodborne illness in humans, representing more than one third of reported outbreaks. After a steady decline between 2005 and 2015, the annual number of human salmonellosis cases increased in France between 2018 and 2022 (European Food Safety et al., 2023). Along with poultry, cattle farming is recognized as a major source of human infections. While the prevalence of *Salmonella* barely exceeds 2 % of animal batches on poultry farms (broilers and layers) (European Food Safety et al., 2023), the intestinal carriage of *Salmonella* was estimated at approximately 3 % in cattle at slaughter (Bonifait et al., 2021).

2. Materials and methods

2.1. Identification of data sources

Following a comprehensive mapping of the food chain surveillance network in France (David et al., 2011; Van Cauteren et al., 2017; Lailler et al., 2020; NOVA, 2020a; Sala et al., 2020), we contacted the

coordinators of the surveillance systems to identify existing data. Three animal health databases were considered relevant for developing and evaluating a SyS approach for human salmonellosis: the National Cattle Register (NCR), the database of the French network for surveillance of antimicrobial resistance in bacterial pathogens isolated from diseased animals (RESAPATH), and the *Salmonella* network database. Our selection was driven by the accessibility and quality of the data, and by the animal species monitored (cattle). In addition, two human health databases were investigated: that of the Coordinated health surveillance of emergency services (the OSCOUR database), and of the General practitioner house call network (the SOS Médecins database).

Each one was split into several subsets (e.g., by syndrome or age group), except for the two human health data sources, which were combined into one subset. In total, 13 subsets were considered and time series were produced for the 2011–2018 period. The description of data sources, associated subsets, and their characteristics are presented in Table 1.

2.2. Data sources and indicators

The NCR, managed by the French Ministry of Agriculture, has been ensuring reliable traceability of cattle in France since 2001. Data are centralized on a daily and mandatory basis. This database includes a variety of information such as bovine identification, breed, age, and farm location. From this database, we created six subsets depicting on-farm mortality rate depending on age (all ages; 0–20 days old; 21 days or older) and cattle production type (beef or dairy). The mortality rate (Table 1) was calculated as the number of deaths divided by the total number of cattle present for the age category under consideration, as previously described (Perrin et al., 2010).

The RESAPATH network, created in 1982, is coordinated by the French Agency for Food, Environmental and Occupational Health & Safety (ANSES). RESAPATH collects antibiogram results from voluntary French veterinary laboratories (Boireau et al., 2018). The antibiograms are requested by veterinarians in a pathological context for diagnostic purposes. The data collected include the identification of isolated bacteria and the pathology of the sampled animal (in a standardized format). Four subsets were created: three syndromic subsets (digestive, mastitis, and reproductive syndromes in cattle, whatever the isolated strains) and one subset specific to *Salmonella* spp. isolations from cattle, whatever the observed syndrome (Table 1).

The *Salmonella* network (<https://www.anses.fr/en/content/salmonella-network>) is a French epidemiological surveillance system that has been monitoring non-human *Salmonella* spp. throughout the food chain since 1997. It is composed of roughly 130 voluntary public and private laboratories and is administered by ANSES. The data recorded consist of *Salmonella* isolations from various ecosystems: livestock farms, slaughterhouses, food processing industries, and food products. We created two subsets covering *Salmonella* isolations in the human food sector (from animal carcasses to food on a plate) and in the livestock sector (from farm environments and animal feed mills).

The OSCOUR and SOS Médecins databases, recorded as part of the French human health SyS system (SURSAUD), are administered by the French public health agency (Santé Publique France). The OSCOUR database records visits to emergency services (and reports the corresponding cause) transmitted voluntarily by more than 600 public emergency services (86 % of all emergency services). The SOS Médecins database collects diagnoses from practitioners made during home visits, communicated on a voluntary basis by 60 local associations of house-call emergency services (96 % of the French SOS Médecins associations). These two networks are complementary: The OSCOUR network covers the whole of France, whereas the SOS Médecins associations are engaged exclusively in urban and semi-urban areas. We created one subset by combining both data sources, depicting the evolution of human gastroenteritis syndromes (number of visits for a gastrointestinal syndrome out of the total number of visits) (Table 1).

Table 1
Description and characteristics of data sources and subsets analyzed.

Dataset (acronym)	Data collection	Centralization		Coverage	Subset	Indicators	
		Frequency	Time taken			Definition	Unit
National cattle register (NCR)	Mandatory	Daily	< 1 day	100 % of national herds	- Beef cattle of all ages - Beef cattle aged < 21 days - Beef cattle aged $\geq$ 21 days - Dairy cattle of all ages - Dairy cattle aged < 21 days - Dairy cattle aged $\geq$ 21 days	Number of deaths divided by the number of cattle-years (for each age group and production type)	Deaths per 100 cattle-years
Surveillance network for antimicrobial resistance in pathogenic bacteria of animal origin (RESAPATH)	Voluntary	Monthly to yearly	Up to 1 year	71 laboratories involved in 2018	- Digestive syndromes - Mastitis syndromes - Reproductive syndromes - <i>Salmonella</i> isolations	- Proportion of antibiograms that are related to digestive pathologies - Proportion of antibiograms that are related to mastitis pathologies - Proportion of antibiograms that are related to reproductive pathologies - Proportion of antibiograms that are related to <i>Salmonella</i> isolations	Proportion
National epidemiological surveillance system of non-human <i>Salmonella</i> throughout the food chain (<i>Salmonella</i> network)	Voluntary	Daily	1 week to 1 year (depends on laboratory)	127 laboratories involved in 2017	- Human food - Livestock	- Number of <i>Salmonella</i> isolations from the human food sector - Number of <i>Salmonella</i> isolations from the livestock sector	Count
Coordinated health surveillance of emergency services (OSCOUR)	Voluntary	Daily	< 1 day	86 % of national visits to emergency services in 2015	SURSAUD (Human gastroenteritis syndromes)	Proportion of total admissions + consultations related to gastroenteritis syndromes	Proportion
General practitioner house call network (SOS Médecins)	Voluntary	Daily	< 1 day	90 % of national at-home emergency visits in 2015			

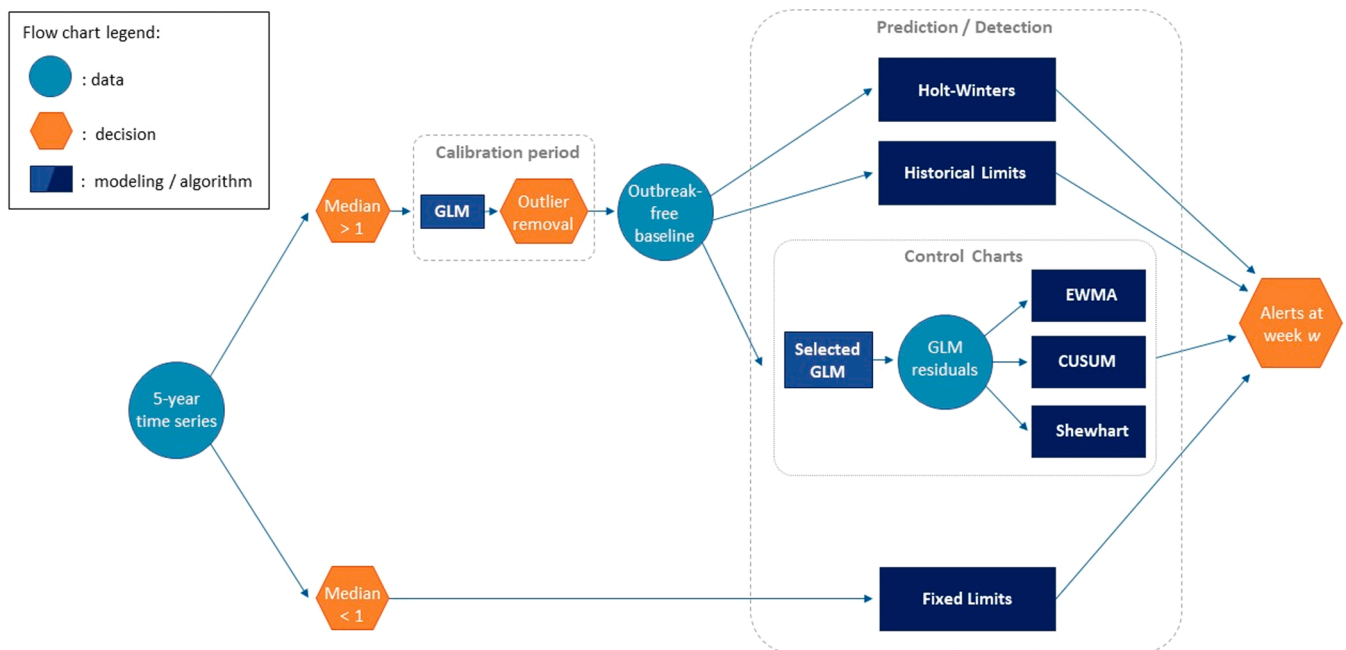


Fig. 1. Flow chart summarizing the different steps of the anomaly detection process for each evaluated week (w) and each time series. CUSUM: cumulative sum control chart, EWMA: exponential weighted moving average control chart, GLM: generalized linear model.

Data analyses were conducted at both national level (i.e., continental France) and regional level, focusing on the French Rhône-Alpes region (in central-eastern France); the latter is referred to as a 'case study' hereafter. This French region was selected because of its production of raw milk cheese, which is a typical carrier for foodborne outbreaks (Ung et al., 2019). We solicited ANSES and the local Livestock Health Protection Groups (GDS) for records of epidemiological events concerning on-field *Salmonella* spp. isolations from animals and animal-based products, and suspicions of *Salmonella* infection in humans in this region in 2018. For each epidemiological event, we considered the date of occurrence. In 2018, the GDS reported *Salmonella* spp.-related events in 13 different weeks, and ANSES reported two clusters of human salmonellosis in September and October (five consecutive weeks)

2.3. Time-series analysis

Data in each subset were aggregated by week (ISO format) at the national scale, and at the regional scale for the case study. For each subset, we tested each of the weeks from week 16 of 2016 to week 52 of 2018 (140 five-year time series). The time series was divided into a calibration period (week 1 to week 260), a guard band of seven weeks (week 261 to week 267), and a test period (week 268).

For each time series, we implemented a classical approach consisting in: i) preparing the calibration period to get a clean baseline, ii) predicting the expected value for the test week, and iii) conducting a retrospective anomaly detection analysis to detect a potential excess event (anomaly) by comparing observed and predicted values during the test period (Fig. 1) (Dorea et al., 2015; Sala et al., 2020).

The cleaned baseline of each time series was obtained using a generalized linear model (GLM) adjusted for over-dispersion if required, testing Poisson, generalized Poisson, quasi-Poisson, and negative binomial distributions. We also tested trend (none and linear), seasonality (annual, bi-annually), and auto-regressive components over the last four weeks. The best-fitting model was selected using the Akaike information criterion (AIC) (Wei, 2006) while the best-fitting distribution was determined using the coefficient of dispersion and an assessment of residual plots. The cleaned baseline was obtained by replacing the values observed above the 95 % confidence interval (CI) by their respective 95 % CI upper limit of the fitted value (Tsui et al., 2001).

Five detection algorithms (Fig. 1) were tested: The historical limits method (HLM) (Stroup et al., 1989; Levin-Rector et al., 2015) and the Holt-Winters algorithm (Gelper et al., 2010; Dorea et al., 2013a) were applied directly to the cleaned baseline, while three control charts (EWMA, CUSUM and Shewhart) were applied to the GLM. These five algorithms were parametrized as previously described (Sala et al., 2020). For Holt-Winters and historical limits algorithms, we compared the observed value with seven threshold values, defined as the upper confidence limit of seven confidence intervals. We used seven values from 1.65 to 3.25 standard deviations to obtain the seven thresholds. For the three control charts, seven values from 2.33 to 3.75 standard deviations were used to obtain seven thresholds. (Table 2). This led to a graduation of the excesses detected from 0 (no excess) to 7 depending on the number of thresholds (0–7) exceeded (Dorea et al., 2013b; Sala et al., 2020).

Neither cleaning nor a detection algorithm was applied to time series with a majority of zeros (median under one) because the models did not converge. In such cases, the observed value was directly compared with fixed values of the number of cases ranging from 2 to 9, with a similar graduation for excesses from 0 (no excess) to 7. Each graduated excess is denoted as an alert score hereafter.

3. Results

3.1. Descriptive analysis

The 13 subsets revealed a wide range of characteristics in terms of

Table 2

Alert scores ranked according to the number of thresholds strictly exceeded. Thresholds are based on upper limit of confidence intervals for control charts, historical limits and Holt-Winters methods or based on an arbitrary threshold (corresponding to a number of cases) for the fixed limits approach. CUSUM: cumulative sum control chart, EWMA: exponential weighted moving average control chart.

Alert severity	Control charts (CUSUM, EWMA, Shewhart)	Historical limits and Holt-Winters	Fixed limits
Threshold 1	2.33σ	1.65σ	2
Threshold 2	2.58σ	1.96σ	4
Threshold 3	2.75σ	2.33σ	5
Threshold 4	3.00σ	2.58σ	6
Threshold 5	3.25σ	2.75σ	7
Threshold 6	3.50σ	3.00σ	8
Threshold 7	3.75σ	3.25σ	9

seasonality patterns and variability between weeks. Considering cattle mortality (NCR) and human health (SURSAUD) data, the time series depicted regular patterns with distinct seasonality (annual or bi-annual) and limited noise. In contrast, the time series from laboratory analyses (RESAPATH and *Salmonella* network data) showed annual seasonality and high noise (Fig. 1).

3.2. Anomaly detection at national level

In the 13 studied subsets, we identified alerts for at least one week out of the 140 weeks tested: Alerts occurred for 32 % (45/140) of the tested weeks for the *Salmonella* network subsets, 43 % for the SURSAUD human health subsets (60/140), between 24 % and 28 % for the RESAPATH network subsets (34–39 /140), between 27 % and 35 % for the NCR dairy subsets (38–49/140), and between 16 % and 28 % for the NCR beef subsets (23–39/140) (Table 3).

We found differences in the number of alerts produced depending on the algorithm. For example, of the 45 alerts in the *Salmonella* Human food subsets, 37 were triggered by the historical limits algorithm (i.e. 82 %) and 22 by CUSUM (i.e. 49 %). The HLM triggered the most alerts in almost all subsets except SURSAUD and the NCR subsets for dairy and beef cattle under 21 days, where the EWMA algorithm triggered the most. For each subset, the number and proportion of alerts triggered by each method are described in Table 3.

We found slight differences in the number of weeks with excesses between the NCR subsets. Overall, the NCR subsets including all age classes and the two NCR subsets including one age class (under 21 days or above 21 days) predicted the same results (i.e., either no alert or alert) for 92 % of the weeks in beef cattle and 77 % of the weeks in dairy cattle.

The characteristics of the alerts in terms of duration, intensity (i.e., score), and the algorithms involved varied greatly between subsets (Fig. 2). Some alerts were reported simultaneously by multiple algorithms across datasets. For the *Salmonella* network, alerts occurred annually in September-October (for a duration of about 10 weeks), with a higher cumulated score for the human food subset (meaning that more algorithms detected these excesses than the ones in the animal livestock subset). For the RESAPATH network, several alerts occurred simultaneously in two subsets: reproduction and mastitis subsets for weeks 27–34 of 2016; and reproduction and *Salmonella* subsets for weeks 32–40 of 2017. Several excesses were detected simultaneously and on consecutive weeks in both the human health subset and the laboratory subset (RESAPATH and *Salmonella* network) on weeks 42–47 of 2016 and 32–38 of 2018. Similarly, simultaneous alerts occurred in the NCR and human health subsets, particularly in weeks 18–20 and 45–50 of 2016.

Overall, for an alert observed in a given week in the human health subset, the proportion of weeks with an alert status in at least two other subsets was 75 % without a delay (i.e., the same week). This percentage

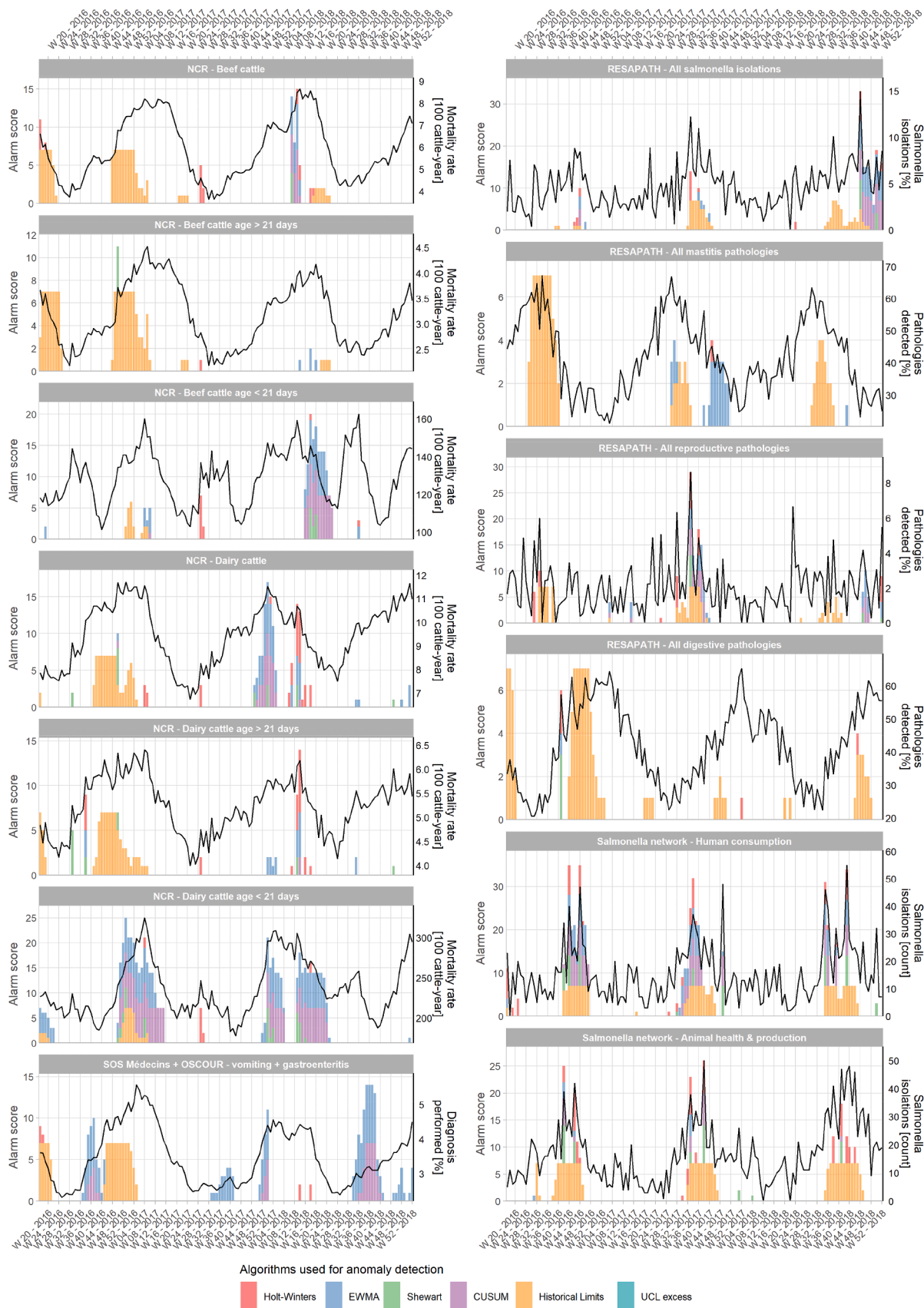


Fig. 2. Observed weekly data (black line) and model-predicted anomalies for the 13 subsets and each algorithm over the 140 weeks tested at a national scale. Cumulative alert scores range from 0 to 7 per algorithm. The left y-axis shows the cumulated alert scores, with colors corresponding to each different algorithm (0–7 thresholds per algorithm). The right y-axis shows the observation units.

was 77 % with an earlier delay of 1–2 weeks (i.e., 1–2 weeks before the human health excess event) and 75 % with an earlier delay of 3–4 weeks. The proportion was 83 % for a later delay of 1–2 weeks (i.e., 1–2 weeks after the human health excess event).

3.3. Case study

The fixed limit approach was required for four subsets for which the average number of observations was very low (median lower than one). Overall, excess events were identified in the Rhône-Alpes case study in each subsets except the NCR subset for dairy cattle aged ≥ 21 days (Fig. 3).

The results showed some overlaps in the weeks when alerts occurred in several subsets. Several alerts ($n = 14$) occurred simultaneously in the two subsets of the *Salmonella* network, for example. Similarly, several alerts ($n = 7$) occurred simultaneously in the reproduction and *Salmonella* subsets of the RESAPATH network. Among all these alerts, three were detected simultaneously in the human health dataset and the laboratory datasets (RESAPATH and *Salmonella* network) on weeks 32 and 35–36 of 2018.

In early 2018, the animal health and production subset of the *Salmonella* network was marked by alerts over five consecutive weeks (weeks 2–6), followed by alerts in the human health dataset (weeks 9–14).

Some excess events occurred at the same time as or just before events registered by ANSES and the local GDS (represented by shaded areas in Fig. 3). Regarding the events reported for weeks 27–40, the animal health and production subset of the *Salmonella* network triggered an alert one week earlier (week 26); the RESAPATH *Salmonella*, RESAPATH reproduction and the human subsets triggered eight, ten and three alerts, respectively, concurrent with these events. Concerning the event reported by the GDS for week 47, we note an alert i) for the week before in all RESAPATH subsets, ii) for the two weeks before in the human food subset of the *Salmonella* network, and iii) the following week in the human health dataset.

4. Discussion

This analysis focuses on the surveillance and prevention of human

salmonellosis through a multifaceted SyS of *Salmonella* contaminations along the food chain. This approach was designed around multiple datasets (from both animal and human health sectors) to which we applied a combination of algorithms to enhance detection timeliness and sensitivity.

Over 20 data sources were identified by mapping food chain surveillance in France (NOVA, 2020a). Data sources were rigorously selected based on their relevance for human salmonellosis SyS and their data quality. Although poultry and pig productions are also known reservoirs of *Salmonella* spp., we were not able to include these species in the study as we did not obtain access to the results of *Salmonella* surveillance programs for either production at the time of the study.

It would be beneficial to incorporate data from these animal species in the study so as to improve the ability of the modeling approach to detect *Salmonella* outbreaks before they reach human populations. Nevertheless, our study focused on the cattle sector, which is another recognized reservoir of *Salmonella* that is likely underestimated in France (Bonifait et al., 2021). In the human sector, contamination is a recurring public health concern and mainly occurs through the consumption of raw milk products and raw or undercooked meat (Vaillant et al., 1996; De Valk et al., 2000; De Buyser et al., 2001; Ung et al., 2019). In parallel, data quality is a key issue when attempting to develop a reliable real-time SyS (Sosin, 2003; Pascal et al., 2012); in our study, coarse temporal resolution, inconsistent data collection, and short data history were major constraints when selecting the data sources. However, the sources we selected had already been checked and completed by those in charge of data collection, so these particular issues did not affect their use in this study (NOVA, 2020b). Data confidentiality, for political or economic reasons, was another issue limiting data access and use. Such restrictions can have a negative impact on the completeness, timeliness, and quality of surveillance data, thus affecting the sensitivity of SyS systems in detecting outbreaks or revealing trends. To address data confidentiality issues, it is essential to establish partnerships and build trust, for example by relying on data-sharing agreements or creating multi-disciplinary expert groups in charge of data analysis and interpretation.

When developing our indicators, whenever possible we took into account the baseline population through a denominator, because changes in surveillance modalities (e.g., volumes or frequency of

Table 3

Number of weeks with alerts according to the type of algorithm that originated the alert on a national scale. CUSUM: cumulative sum control chart, EWMA: exponential weighted moving average control chart, NCR: national cattle register.

Subsets	Number (percentage) of weeks with at least one algorithm in alert status over the 140 evaluated weeks	Number (percentage) of weeks in alert status out of the total number of alerts detected for each algorithm tested				
		Holt-Winters	EWMA	CUSUM	Shewart	Historical limits
<i>Salmonella</i> Human food	45 (32 %)	16 (36 %)	22 (49 %)	22 (49 %)	9 (20 %)	37 (82 %)
<i>Salmonella</i> Livestock	45 (32 %)	14 (31 %)	4 (9 %)	5 (11 %)	7 (16 %)	41 (91 %)
SURSAUD (Human health)	60 (43 %)	4 (7 %)	41 (68 %)	17 (28 %)	-	18 (30 %)
RESAPATH <i>Salmonella</i>	39 (28 %)	9 (23 %)	15 (38 %)	10 (26 %)	2 (5 %)	32 (82 %)
RESAPATH Mastitis	39 (28 %)	1 (3 %)	13 (33 %)	-	-	29 (74 %)
RESAPATH Reproductive	34 (24 %)	7 (21 %)	13 (38 %)	10 (29 %)	3 (9 %)	25 (74 %)
RESAPATH Digestive	38 (27 %)	3 (8 %)	1 (3 %)	-	1 (3 %)	36 (95 %)
NCR Beef all	39 (28 %)	8 (21 %)	4 (10 %)	4 (10 %)	1 (3 %)	32 (82 %)
NCR Beef ≥ 21 days	35 (25 %)	1 (3 %)	3 (9 %)	-	1 (3 %)	31 (89 %)
NCR Beef < 21 days	23 (16 %)	4 (17 %)	14 (61 %)	12 (52 %)	3 (13 %)	7 (30 %)
NCR Dairy all	43 (31 %)	11 (26 %)	17 (40 %)	10 (23 %)	6 (14 %)	18 (42 %)
NCR Dairy ≥ 21 days	38 (27 %)	7 (18 %)	8 (21 %)	1 (3 %)	5 (13 %)	24 (63 %)
NCR Dairy < 21 days	49 (35 %)	4 (8 %)	44 (90 %)	36 (73 %)	11 (22 %)	15 (31 %)

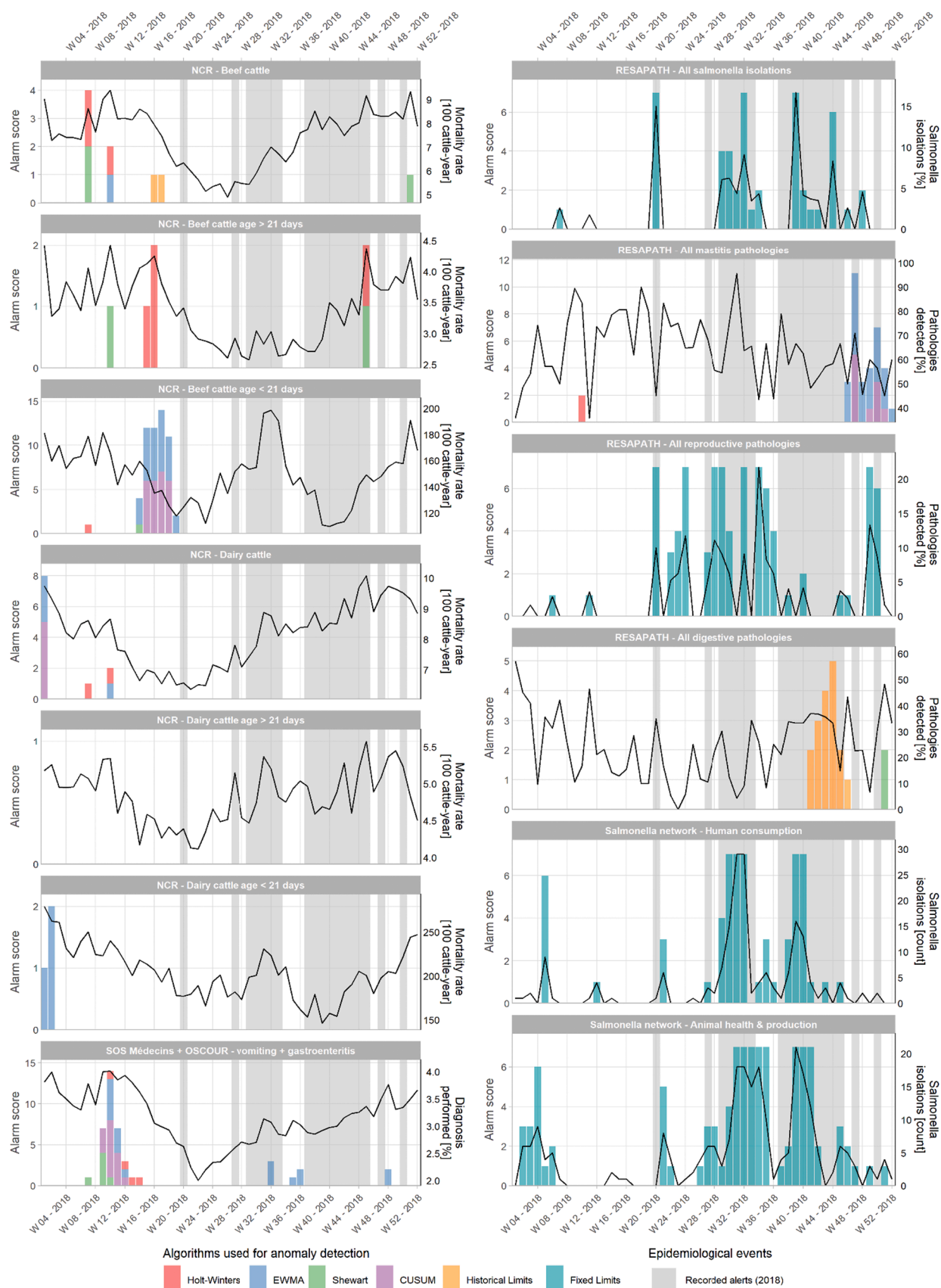


Fig. 3. Observed weekly data (black line) and model-predicted anomalies for the 13 subsets and each algorithm over the 52 weeks of 2018 tested in the Rhône-Alpes region. Cumulative alert scores range from 0 to 7 per algorithm. The left y-axis shows the cumulated alert scores, with colors corresponding to each different algorithm (0–7 thresholds per algorithm). The right y-axis shows the observation units.

sampling) or at-risk populations have a strong influence on the anomalies detected in time series. However, data on populations covered by surveillance may be scarce and difficult to obtain for real-time monitoring. In such cases, a proxy can be used to replace this missing information such as the total number of tests/consultations, as used in the Human SyS of *Santé Publique France*.

The frequency of data aggregation plays a crucial role in real-time monitoring. Two scales may be appropriate: daily or weekly. Daily aggregation needs a well-structured organization for data centralization and interpretation, and requires a sufficient number of daily events for time series modeling. Weekly aggregation is more appropriate for limiting statistically random variation due to a small daily number of events. The weekly time interval is compatible with the response to field realities regarding zoonotic disease dynamics. ISO week date (Monday to Sunday) aggregation is usually applied in epidemiology studies. We also adopted this weekly scale to bypass the weekday effect observed in some daily time series (Perrin et al., 2010; Brilleaud et al., 2022). Nevertheless, the main limit of this process is that small excesses over a few days may be hidden by the aggregation, increasing the risk of non-detection. An alternative mixed solution could be to have daily analyses of data aggregated by sliding week. However, depending on diseases or syndromes selected for the analyses, daily counts may be insufficient for analyses at a daily scale (Dorea et al., 2013c). In addition, daily data are rarely available in the veterinary sector because veterinary and laboratory practices are generally independent and each has its own management software. The veterinary sector may also have less stringent reporting requirements than the human health sector, particularly for non-regulatory diseases. Moreover, while data analyses can be automated, this solution requires upstream daily data collection and recording in the database, and daily interpretation of alerts and investigations when relevant. Because of limited technical and human resources, an appropriate balance between sensitivity and specificity must be found by calibrating a maximum number or rate of false alerts that can be investigated by field actors. If staff are unavailable or overloaded, alerts may not be investigated or responses may be delayed, reducing the effectiveness of the system in preventing epidemics.

Our study demonstrated the potential of the selected data sources for *Salmonella* SyS at both national and regional scales. However, the choice of geographic resolution is highly dependent on the phenomena to be covered. Indeed, while sometimes product consumption is local (over the counter or directly at the farm), the vast majority of food products come from many different processing plants that distribute their products nationwide. Thus, the geographic locality of a human case can be a long way from the place of contamination, making the interpretation of alerts more difficult. Therefore, analysis on a national scale would be appropriate for detecting a widespread, low-level spread of *Salmonella*, and regional analysis would be appropriate for detecting specific events related to the spatial distribution of the cattle population of interest or food products. It is important to stress that the choice of geographic resolution influences SyS sensitivity: when considering a large spatial unit (such as a country), local outbreaks may be diluted in the dataset (Ozonoff et al., 2007), whereas they would be detected if a smaller spatial unit (such as regions) was used, especially when the at-risk population is homogeneously spatially distributed. Nonetheless, it is only possible to use a small geographic unit if there are enough data to avoid poor statistical robustness related to the “small numbers problem” (Pringle, 1996).

To increase the specificity of anomaly detection and help identify signals that require field investigation, two complementary approaches were tested: (i) the use of a combination of algorithms, assuming that highly consistent results between the different algorithms indicate high likelihood of a signal; and (ii) the use of multiple detection limits to increase the specificity (high threshold) while maintaining sensitivity (low threshold). We implemented five algorithms with seven thresholds to fully grasp the diversity of monitoring data. We selected widely accepted algorithms, already implemented in existing SyS systems

(Dorea and Vial, 2016; Sala et al., 2020) from a wide range of previously reviewed algorithms (Faverjon and Berezowski, 2018). The combination of algorithms and thresholds provides an initial assessment of excess events identified (nature, severity) before field investigation. One of the major challenges of SyS is to summarize results in such a way as to ensure a clear understanding of alerts for epidemiologists and decision makers alike. By considering multi-source data, the graphs (e.g., Fig. 2) and tables (e.g., Table 3) produced enable a detailed interpretation of the temporality of the epidemics studied. This interpretation differs from what would have been detected (or not) if only human data or only animal data had been monitored. They reveal simultaneous or staggered alerts when comparing data sources but do not justify a causal relationship. Only epidemiological investigations can confirm the occurrence of a specific event. This is not covered in our study, although the Rhône-Alpes case study is a first step in this direction. Indeed, such a system could help decision makers investigating human cases by considering animal commodities or alerts triggered along the food chain as potential sources of infection. Furthermore, the detection of concomitant or sequentially close events in time series based on animal or food-chain data allows the implementation of control measures in these sectors, together with preventive measures to limit the pathogen's spread to humans. Collaboration with field investigators is particularly important to minimize false alerts (Smith et al., 2019a), just as the commitment of stakeholders and data reporters is essential to the long-term success of any SyS system (Pfeiffer et al., 2021). It is essential to strengthen the commitment and collaboration between academic researchers, government agencies and private partners (such as farmers or veterinarians), who can operationalize such surveillance systems and ensure their long-term funding and sustainability.

Furthermore, whole genome sequencing (WGS) is currently a powerful method to usefully estimate the contributions of different food categories or animal species as sources of human infections (Munck et al., 2020). By comparing genomes, WGS reveals the strength of the epidemiological relationship between two strains. For *Salmonella* surveillance, very rare or specific serovars isolated from a given source would be useful for assessing sensitivity because of the higher probability of detecting related strains (a true signal) harboring similar genomes (Ung et al., 2019). However, despite technological advances, both the processing of WGS data and their interpretation require specialized software and bioinformatics skills, and are therefore relatively expensive, especially when applied to large-scale routine surveillance. Besides, the sharing of (WGS) data requires adherence to FAIR principles (Wilkinson et al., 2016) and the implementation of specific tools and data-sharing agreements to respect confidentiality issues. *Salmonella* surveillance in France is currently based on different systems (event-based and programmed surveillance), the data of which are analyzed separately. Our proof-of-concept study has demonstrated the advantages of developing a SyS system that could make surveillance more efficient, as has been shown in animal health by The French National Animal Health Surveillance Platform (Dupuy et al., 2024). This study is therefore an essential step in the operational development of such a system. Another promising research opportunity would be to focus on multivariate analyses, considering dependency relationships between various surveillance data subsets. In a One Health approach, interaction between the medical, veterinary, and environmental domains is necessary to protect public and animal health. For this kind of collaboration to work successfully, it is advisable to build such systems in ‘peacetime’, so that they are ready in the event of a crisis. Accordingly, the One Health European Joint Project COHESIVE has developed a web-based implementation guideline to support countries wishing to set up an operational One Health risk analysis system (www.ohras.eu).

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Ethical statement

Ethical approval was not needed for this study because it is based on aggregated, anonymous surveillance data.

CRedit authorship contribution statement

Briac Virey: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis. **Géraldine Cazeau:** Writing – review & editing, Writing – original draft, Software, Resources, Methodology, Formal analysis. **Renaud Lailler:** Writing – review & editing, Resources. **Carole Sala:** Writing – review & editing, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Viviane Hénaux:** Writing – review & editing, Resources, Project administration, Funding acquisition, Conceptualization. **Adeline Huneau-Salaün:** Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of Competing Interest

None declared.

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